

GROUP PROJECT GUIDELINE

TOPIC 2: RETAIL AND ECOMMERCE – RECOMMENDATION ENGINE FOR PERSONALIZED SHOPPING

OVERVIEW

Course: TMA Advanced AI Training

Type: Group Projects (10-15 members per group)

Objectives: This project aims to provide learners with hands-on experience in developing a complete AI-driven recommendation system tailored for the retail and e-commerce sector. By leveraging publicly available datasets on user behavior, product metadata, and purchase history, students will explore the full development lifecycle—from data collection, preprocessing, and feature engineering, to building and training machine learning models for personalized product recommendations. The project will also include evaluation of model performance using metrics like precision, recall, and F1-score, as well as deployment of the solution through an interactive web interface. The ultimate goal is to create a recommendation engine that enhances the online shopping experience by suggesting relevant products to users, thereby increasing engagement, improving customer satisfaction, and boosting sales conversion rates.

DATASET

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

No.	Name	Source	Description
1	Retailrocket recommender system dataset	Link	Includes user behavior data (clicks, add to cart, transactions) along with product and category information.
2	Amazon Review dataset	Link	This is a large-scale Amazon Reviews dataset
3	eCommerce behavior data from multi category store	Link	This dataset contains 285 mil users' events from eCommerce website
4	User behavior data on Taobao App	Link	This dataset is provided by Alibaba Group
5	MovieLens dataset	Link	MovieLens with 32 mil movie ratings

GUIDELINES

Each group will submit 1 report with the following constraints:

- Report must include reference/citation of the sources being used
- Visualization can be included in the report but must not exceed 50% of the total space
- Report can range from 10 to 15 pages

REPORT GUIDELINE

Your final report should be structured according to the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** standard ([Link](#)) and should include the following sections:

- 1. Business Understanding:**
 - Clearly define the problem your recommendation engine will solve
 - Identify business goals (e.g., boost sales, improve user experience)
 - Understand pain points: product overload, irrelevant suggestions, low conversions
 - Set success criteria (e.g., improved click-through rate, higher engagement)
- 2. Data Understanding:**
 - Explore the dataset(s): structure, features, and source
 - Identify key data points: user ID, item ID, timestamps, events (click, purchase)
 - Look for patterns: top products, frequent users, seasonal trends
 - Note external factors: holidays, marketing campaigns, price changes
- 3. Data Preparation:**
 - Clean the data: handle missing values, duplicates, and outliers
 - Convert timestamps into meaningful formats (e.g., sessions, days, weeks)
 - Create useful features: product popularity, user preferences, recency of actions
 - Add flags: holidays, special events, or user segments
- 4. Modeling:**
 - Choose model types:
 - Collaborative filtering (e.g., Matrix Factorization)
 - Content-based filtering
 - Hybrid models
 - Prepare training and testing data
 - Apply hyperparameter tuning to improve accuracy
 - Ensure your model is scalable and efficient
- 5. Evaluation:**
 - Use standard metrics (e.g. Precision, Recall, F1-score)
 - Compare different models and baselines
 - Identify where your model performs well or needs improvement
- 6. Deployment/Delivery:**
 - Build a simple app, dashboard, or interface to show recommendations
 - Visualize results and user flows
 - Summarize insights and real-world impact

- Suggest what could be added next (e.g. Real-time updates, Personalization with user profiles, Image or text-based recommendations)

GRADING GUIDELINE

No.	Name	Description	Grading Percentage
1	Data	Clearly defined data sources and collection methods	30%
2	AI/ML Technology	Explanation of AI model, techniques and implementation	40%
3	Demo	Functional code demonstrating the AI solution	20%
4	Reference & Citation	At least 5 credible sources supporting the project	10%